ModelIt!

Teacher's Guide

Why Some People Can't Drink Milk

Gene Expression, Enzymes, and the Lactose Puzzle

Quick Reference

Grade Level	Grade 09L1
NGSS Standard	HS-LS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS3-1

Ask questions to clarify relationships about the role of DNA and chromosomes in coding the instructions for characteristic traits passed from parents to offspring.

Three-Dimensional Learning

SEP	**Asking Questions and Defining Problems** — Students ask questions about how DNA sequence variations lead to different enzyme production levels and downstream phenotypic differences across populations.
DCI	**LS3.A: Inheritance of Traits** — DNA codes for proteins (enzymes). Variations in gene regulation determine how much lactase enzyme is produced, directly causing the observable trait of lactose tolerance or intolerance. This is gene expression controlling phenotype.
CCC	**Cause and Effect** — A clear causal chain connects gene activity → enzyme production → substrate breakdown → symptom presence. Changing the initial cause (gene expression level) predictably changes the final effect (digestive outcome).

CAST Testing Alignment

EXPLAIN:

Two siblings eat identical ice cream sundaes. One feels fine; the other has severe stomach cramps an hour later. Using your model, trace the causal chain from their LCT gene activity to their different outcomes.

ANALYZE:

About 90% of East Asian populations are lactose intolerant while 90% of Northern Europeans are tolerant. Using your understanding of gene expression and natural selection, explain this difference.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Lactose intolerance is one of the most elegant examples of how a single gene controls a physical trait through enzyme production — and how human culture (dairy farming) drove genetic evolution in real time.

The Lactase Gene (LCT)

The LCT gene on chromosome 2 encodes the enzyme lactase-phlorizin hydrolase, which sits on the brush border of small intestine epithelial cells. In all mammals, this gene is active during infancy (when the sole food is mother's milk) and naturally decreases in expression after weaning. This 'lactase non-persistence' is the ancestral, default human condition — NOT a disorder.

The Lactase Persistence Mutation

About 7,500 years ago, a mutation arose upstream of the LCT gene that prevents the normal age-related decrease in expression. This mutation (specifically, a C-to-T change at position -13910) doesn't change the enzyme itself — it changes REGULATION, keeping the gene 'turned on' into adulthood. This mutation spread rapidly in populations that domesticated dairy cattle, providing a powerful selective advantage: access to a rich, reliable food source.

Why Symptoms Occur

When lactose reaches the large intestine undigested, resident bacteria eagerly ferment it through anaerobic metabolism. This produces hydrogen gas, methane, and short-chain fatty acids. The gas causes bloating and cramps. Additionally, the undigested lactose is osmotically active — it draws water into the intestinal lumen through osmosis, causing diarrhea. The severity depends on how much lactose reaches the colon, which depends on how much lactase enzyme was available.

Global Distribution

Lactase persistence follows dairy farming history almost perfectly. Northern European populations (90%+ tolerant), East African pastoralist groups like the Maasai and Tutsi (50-90% tolerant), and some Middle Eastern populations show high tolerance — all cultures with thousands of years of dairy tradition. East Asian, West African, Native American, and Aboriginal Australian populations show 70-100% intolerance — cultures without historical dairy farming. This is one of the clearest examples of gene-culture co-evolution.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify lactase gene activity, enzyme production, lactose breakdown, and gut bacteria response as components.

E **Establish Relationships**
Students discover the causal chain: gene activity (+) enzyme → enzyme (+) breakdown → undigested lactose (+) bacteria response.

V **Visualize & Model**
Students build a model showing how gene expression level determines digestive outcomes.

E **Evaluate Outcomes**
Students run tolerant, intolerant, and partial scenarios to discover threshold effects.

R **Revise & Extend**
Students add dairy amount, microbiome composition, or epigenetic regulation for more realistic modeling.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Raise your hand if ice cream has ever made you feel terrible. Now raise your hand if it never has. Why the difference?"*

Watch For

Quick informal poll. Don't ask students to self-identify — just note the split exists.

Slide 2: Learning Objectives

3 min

Say: *"Today we're solving a genetics mystery: why your DNA decides whether you can drink milk."*

Watch For

Have students read objectives. Pre-teach 'gene expression' as 'reading a recipe from the cookbook.'

Slide 3: Big Question

3 min

Say: *"Same milkshake, two people, completely different outcomes. One gene makes the difference."*

Watch For

Show the global map of lactose tolerance. Ask: Why the geographic pattern?

Slide 4: LEVER Framework

2 min

Say: *"We'll model how ONE gene controls whether you can enjoy pizza or suffer after it."*

Watch For

Preview LEVER steps. Emphasize the causal chain: gene → enzyme → digestion → symptoms.

Slide 5: Activity 1: Components

8 min

Say: *"Which part of this system is the 'input' and which parts are 'outputs'?"*

Watch For

Guide sorting. Gene activity is the key external driver — everything else cascades from it.

Slide 6: Activity 2: Connections

8 min

Say: *"More gene activity means more enzyme. More enzyme means more breakdown. But what happens when there's NOT enough enzyme?"*

Watch For

Draw the complete causal chain. Show how it branches at 'undigested lactose → bacteria.'

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test three genetic scenarios: fully tolerant, fully intolerant, and somewhere in between."*

Watch For

Students predict outcomes. The partial tolerance threshold is the key discovery.

Slide 8: Discoveries

5 min

Say: *"Here's the twist: lactose INTOLERANCE is the default. Tolerance is the mutation. We evolved to stop drinking milk."*

Watch For

Show the evolutionary timeline: mutation arose 7,500 years ago with dairy farming.

Slide 9: STEM Challenge

12 min

Say: *"65% of the world can't drink milk. Design a product that solves this problem using enzyme science."*

Watch For

Groups design solutions: pre-added lactase, bacterial fermentation, or dosing strategies.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Lactase Gene Activity
- **INTERNAL:** Lactase Enzyme Production, Lactose Breakdown, Gut Bacteria Response

Relationships

- **Lactase Gene Activity to Lactase Enzyme Production:** POSITIVE (+) — Higher gene expression means more mRNA is transcribed and translated into functional lactase protein. The gene is the blueprint; more reading = more enzyme.
- **Lactase Enzyme Production to Lactose Breakdown:** POSITIVE (+) — More enzyme molecules in the small intestine means more lactose molecules get broken down into absorbable glucose and galactose before reaching the large intestine.
- **Lactose Breakdown to Gut Bacteria Response:** NEGATIVE (-) — When more lactose is broken down in the small intestine, less undigested lactose reaches the large intestine bacteria. Higher breakdown means less fermentation and fewer symptoms.

Key Discoveries

- Lactase gene activity directly controls enzyme production, which determines whether lactose gets digested or fermented
- Most humans naturally reduce lactase production after weaning — lactose TOLERANCE is actually the mutation, not intolerance
- There's a threshold effect: some lactase is enough to digest small amounts of dairy, but large amounts overwhelm limited enzyme
- Gut bacteria fermentation of undigested lactose produces gas and osmotic effects — this is what causes the symptoms

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Lactose intolerance is an allergy"	Lactose intolerance is NOT an allergy. Allergies involve the immune system attacking a protein (like milk protein allergy, which is different and rarer). Lactose intolerance is a deficiency — not enough enzyme to break down a sugar. No immune response is involved. The distinction matters because allergies can be life-threatening (anaphylaxis) while lactose intolerance is uncomfortable but not dangerous.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Lactose intolerant people can't eat ANY dairy"	Most lactose-intolerant people can handle some dairy. Aged cheeses (cheddar, parmesan) contain almost no lactose because bacteria consumed it during aging. Yogurt has 20-50% less lactose than milk for the same reason. Many intolerant people can tolerate a splash of milk in coffee. The threshold varies by individual.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Lactose intolerance is a modern disease"	Lactose intolerance is the ancestral human condition — every human before 7,500 years ago was lactose intolerant as an adult. Tolerance is the modern mutation. We've medicalized the DEFAULT state and treated the mutation as normal, probably because medical science was developed primarily in Northern European populations where tolerance is common.